Discrete Mathematics

Combinatorics — Spring 2025

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§1 Combinatorics

Introduction to Combinatorics

Definition 1: Combinatorics is the branch of discrete mathematics that deals with *counting*, *arranging*, and analyzing *discrete structures*.

Three basic problems of Combinatorics:

1. Existence: *Is there at least one arrangement of a particular kind?*
2. Counting: *How many arrangements are there?*
3. Optimization: *Which one is best according to some criteria?*

Discrete structures

- Graphs, sets, multisets, sequences, patterns, coverings, partitions...

Enumeration

- Permutations, combinations, inclusion/exclusion, generating functions, recurrence relations...

Algorithms and optimization

- Sorting, eulerian circuits, hamiltonian cycles, planarity testing, graph coloring, spanning trees, shortest paths, network flows, bipartite matchings, chain partitions...

Discrete Structures

We investigate the *building blocks* of combinatorics:

- Sets and multisets
- Sequences and strings
- Arrangements
- Graphs, networks, trees
- Posets and lattices
- Partitions
- Patterns, coverings, designs, configurations
- Schedules, assignments, distributions

Used in data modeling, logic, cryptography, and the design of data structures.

Enumerative Combinatorics

We learn how to count *without explicit listing*:

- Permutations and combinations
- Inclusion–Exclusion Principle
- Set partitions, integer partitions, Stirling numbers, Catalan numbers
- Recurrence relations
- Generating functions

Used in probability theory, complexity theory, coding theory, computational biology.

Algorithmic and Optimization Methods

Combinatorics powers *algorithm design* and complexity analysis:

- Sorting
- Searching
- Eulerian paths and Hamiltonian cycles
- Planarity, colorings, cliques, coverings
- Spanning trees
- Shortest paths
- Network flows
- Bipartite matchings
- Dilworth's theorem, chain and antichain partitions

Used in logistics, scheduling, routing, and complexity optimization.

§2 Basic Counting Principles

Basic Counting Rules

Product rule: If something can happen in n_1 ways, *and* no matter how the first thing happens, a second thing can happen in n_2 ways, then the two things *together* can happen in $n_1 \cdot n_2$ ways.

Sum rule: If one event can occur in n_1 ways and a second event in n_2 (different) ways, then there are $n_1 + n_2$ ways in which *either* the first event *or* the second event can occur (*but not both*).

Addition Principle

Definition 2: We say a finite set S is *partitioned* into *parts* $S_1, \ldots, S_m$ if the parts are pairwise disjoint and their union is S. In other words, $S_i \cap S_j = \emptyset$ for $i \neq j$ and $S_1 \cup S_2 \cup \ldots \cup S_k = S$. In that case:

$$|S| = |S_1| + |S_2| + \ldots + |S_m|$$

Example: Let S be the set of students attending the combinatorics lecture. It can be partitioned into parts S_1 and S_2 where

$$S_1 = \text{set of students that like easy examples.}$$
$$S_2 = \text{set of students that don't like easy examples.}$$

If $|S_1| = 22$ and $|S_2| = 8$, then we can conclude $|S| = |S_1| + |S_2| = 30$.

Multiplication Principle

Definition 3: If S is a finite set that is the *product* of $S_1, \ldots, S_m$, that is, $S = S_1 \times \ldots \times S_m$, then

$$|S| = |S_1| \times \ldots \times |S_m|$$

Example: TODO: example with car plates

Subtraction Principle

Definition 4: Let S be a subset of a finite set T. We define the *complement* of S as $\overline{S} = T \setminus S$. Then

$$\left|\overline{S}\right| = |T| - |S|$$

Example: If T is the set of students studying at KIT and S the set of students studying neither math nor computer science. If we know $|T| = 23905$ and $|S| = 20178$, then we can compute the number $|S|$ of students studying either math or computer science:

$$|S| = |T| - |S| = 23905 - 20178 = 3727$$

Bijection Principle

Definition 5: If S and T are sets, then

$$|S| = |T| \quad \Longleftrightarrow \quad \text{there exists a bijection between } S \text{ and } T$$

Example: Let S be the set of students attending the combinatorics lecture and T the set of homework submissions (unique per student) for the first problem sheet. If the number of students and the number of submissions coincide, then there is a bijection between students and submissions.

Note: The bijection principle works both for *finite* and *infinite* sets.

Pigeonhole Principle

Definition 6: Let $S_1, ..., S_m$ be finite sets that are pairwise disjoint and $|S_1| + |S_2| + ... + |S_m| = n$.

$$\exists i \in \{1, ..., m\} : |S_i| \geq \left\lfloor \frac{n}{m} \right\rfloor \quad \text{and} \quad \exists j \in \{1, ..., m\} : |S_j| \leq \left\lceil \frac{n}{m} \right\rceil$$

Example: Assume there are 5 holes in the wall where pigeons nest. Say there is a set S_i of pigeons nesting in hole i. Assume there are $n = 17$ pigeons in total. Then we know:

- There is some hole with at least $d = 4$ pigeons.
- There is some hole with at most $b = 3$ pigeons.

Double Counting

If we count the same quantity in *two different ways*, then this gives us a (perhaps non-trivial) identity.

Example (Handshaking Lemma): Assume there are n people at a party and everybody will shake hands with everybody else. How many handshakes will occur? We count this number in two ways:

1. Every person shakes $n-1$ hands and there are n people. However, two people are involved in a handshake so if we just multiply $n \cdot (n-1)$, then every handshake is counted twice. The total number of handshakes is therefore $\frac{n \cdot (n-1)}{2}$.
2. We number the people from 1 to n. To avoid counting a handshake twice, we count for person i only the handshakes with persons of lower numbers. Then the total number of handshakes is:

$$\sum_{i=1}^{n}(i-1) = \sum_{i=0}^{\{n-1\}} i = \sum_{i=1}^{n-1} i$$

The identity we obtain is therefore: $\displaystyle\sum_{i=1}^{n-1} i = \frac{n \cdot (n-1)}{2}$

§3 Arrangements, Permutations, Combinations

Ordered Arrangements

Definition 7: Denote by $[n] = \{1, \ldots, n\}$ the set of natural numbers from 1 to n.

Hereinafter, let X be a finite set.

Definition 8: An *ordered arrangement* of n elements of X is a *map* $s : [n] \longrightarrow X$.

- Here, $[n]$ is the *domain* of s, and $s(i)$ is the *image* of $i \in [n]$ under s.
- The set $\{x \in X \mid s(i) = x \text{ for some } i \in [n]\}$ is the *range* of s.

Other common names for ordered arrangements are:

- *string* (or *word*), e.g. "Banana"
- *sequence*, e.g. "0815422372"
- *tuple*, e.g. $(3, 5, 2, 5, 8)$

Example:

i	1	2	3	4	5	6	7
$s(i)$	🦀	🥖	🥖	🫙	🥖	🦀	🫙

Permutations

Definition 9: A *permutation* of X is a *bijective* map $\pi : [n] \longrightarrow X$.

Usually, $X = [n]$, and the set of all permutations of $[n]$ is denoted by S_n.

Example:

i	1	2	3	4	5	6	7
$\pi(i)$	2	7	1	3	5	4	6

Definition 10: *k-permutation* of X is an ordered arrangement of k *distinct* elements of X, that is, an *injective* map $\pi : [k] \longrightarrow X$.

The set of all k-permutations of $X = [n]$ is denoted by $P(n, k)$. In particular, $S_n = P(n, n)$.

TODO: circular permutations

Counting Permutations

Theorem 1: For any natural numbers $0 \leq k \leq n$, we have

$$|P(n,k)| = n \cdot (n-1) \cdot \ldots \cdot (n-k+1) = \frac{n!}{(n-k)!}$$

Proof: A permutation is an injective map $\pi : [k] \longrightarrow [n]$. We count the number of ways to pick such a map, picking the images one after the other. There are n ways to choose $\pi(1)$. Given a value for $\pi(1)$, there are $(n-1)$ ways to choose $\pi(2)$ (since we may not choose $\pi(1)$ again). Continuing like this, there are $(n-i+1)$ ways to pick $\pi(i)$, and the last value we pick is $\pi(k)$ with $(n-k+1)$ possibilities.

Every k-permutation can be constructed like this in *exactly one way*. The total number of k-permutations is therefore given as the product:

$$|P(n,k)| = n \cdot (n-1) \cdot \ldots \cdot (n-k+1) = \frac{n!}{(n-k)!}$$

$\square$

Counting Circular Permutations

Theorem 2: For any natural numbers $0 \leq k \leq n$, we have

$$|P_c(n,k)| = \frac{n!}{k! \cdot (n-k)!}$$

Proof: We doubly count $P(n,k)$:

1. $|P(n,k)| = \frac{n!}{(n-k)!}$ which we proved before.
2. $|P(n,k)| = |P_c(n,k)| \cdot k$ because every equivalence class in $P_c(n,k)$ contains k permutations from $P(n,k)$ since there are k ways to rotate a k-permutation.

From this we get $\frac{n!}{(n-k)!} = |P_c(n,k)| \cdot k$ which implies $|P_c(n,k)| = \frac{n!}{k! \cdot (n-k)!}$. $\square$

Unordered Arrangements

Definition 11: An *unordered arrangement* of k elements of X is a *multiset* $S = \langle X, r \rangle$ of size k.

In a multiset, X is the set of *types*, and for each type $x \in X$, r_x is its *repetition number*.

Example: Let $X = \{$ 👒 , 🦭 , 🐈 , 🎏 , 🌵 $\}$.

- An unordered arrangement of 7 elements could be $S = \{$ 👒 , 👒 , 🦭 , 🐈 , 🐈 , 🐈 , 🌵 $\}^*$.
- The same multiset could be written as $S = \{2$ 👒 , 1 🦭 , 3 🐈 , 0 🎏 , 1 🌵 $\}$.

Subsets

The most important special case of unordered arrangements is where all repetitions are 1, i.e., $r_x = 1$ for all $x \in X$. Then S is simply a *subset* of X, denoted $S \subset X$.

Definition 12: A *k-combination* of X is an unordered arrangement of k *distinct* elements of X.

Note: The more standard term is *subset*. The term "combination" is only used to emphasize the selection process.

The set of all k-subsets of X is denoted $\binom{X}{k} = \{A \subseteq X \mid |A| = k\}$. If $|X| = n$, then

$$\binom{n}{k} := \left|\binom{X}{k}\right|$$

Example: The set of edges in a simple undirected graph consists of 2-subsets of its vertices: $E \subseteq \binom{V}{2}$.

Counting k-Combinations

Theorem 3: For $0 \leq k \leq n$, we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

Proof: $|P(n,k)| = \dfrac{n!}{(n-k)!} = \dbinom{n}{k} \cdot k!$ $\square$

§4 Multisets

Multiset

Definition 13: A *multiset* is a modification of the concept of a set that allows for *repetitions* of its elements. Formally, it is denoted as a pair $M = \langle X, r \rangle$, where X is the *groundset* (the set of *types*) and $r : X \longrightarrow \mathbb{N}_0$ is the *multiplicity function*.

Example: When the multiset is defined by enumeration, it is advisable to use the notation with the star:

$$M = \{a, b, a, a, b\}^* = \{3 \cdot a, 2 \cdot b\} \quad X = \{a, b\} \quad r_a = 3, r_b = 2$$

Example: Prime factorization of a natural number n is a multiset, e.g. $120 = 2^3 \cdot 3^1 \cdot 5^1$.

k-Combinations of a Multiset

Definition 14: Let X be a finite set of types, and let $M = \langle X, r \rangle$ be a finite multiset with repetition numbers $r_1, \ldots, r_{|X|}$. A *k-combination of M* is a multiset $S = \langle X, s \rangle$ with types in X and repetition numbers $s_1, \ldots, s_{|X|}$ such that $s_i \leq r_i$ for all $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

Example: Consider $M = \{2🦫, 1🥤, 3🍉, 1💎\}$.

- $T = \{1🦫, 2🍉\}$ is a 3-combination of M.
- $T' = \{3💎\}$ is not.

Counting k-combinations of a multiset is not as simple as it might seem...

k-Permutations of a Multiset

Definition 15: Let M be a finite multiset with set of types X. A *k-permutation of M* is an ordered arrangement of k elements of M where different orderings of elements of the same type are *not distinguished*. This is an ordered multiset with types in X and repetition numbers $s_1, \ldots, s_{|X|}$ such that $s_i \leq r_i$ for all $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

Note: There might be several elements of the same type compared to a permutation of a set (where each repetition number equals 1).

Example: Let $M = \{2🦫, 1🥤, 3🍉, 1💎\}$, then $T = (💎, 🍉, 🍉, 🦫)$ is a 4-permutation of multiset M.

Binomial Theorem

Theorem 4: The expansion of any non-negative integer power n of the binomial $(x+y)$ is a sum

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} \cdot x^k \cdot y^{n-k}$$

where each $\binom{n}{k}$ is a positive integer known as a *binomial coefficient*, defined as

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!} = \frac{n(n-1)(n-2)\ldots(n-k+1)}{k(k-1)(k-2)\ldots \cdot 2 \cdot 1}$$

Multinomial Theorem

Theorem 5: The generalization of the binomial theorem:

$$(x_1 + \ldots + x_r)^n = \sum_{\substack{0 \le k_1, \ldots, k_r \le n \\ k_1 + \ldots + k_r = n}}^{n} \binom{n}{k_1, \ldots, k_r} \cdot x_1^{k_1} \cdot \ldots \cdot x_r^{k_r}$$

Multinomial coefficients are defined as

$$\binom{n}{k_1, \ldots, k_r} = \frac{n!}{k_1! \cdot \ldots \cdot k_r!}$$

Note: Binomial coefficients are special cases of multinomial coefficients ($r = 2$):

$$\binom{n}{k} = \binom{n}{k_1, k_2} = \binom{n}{k, n-k} = \frac{n!}{k! \cdot (n-k)!}$$

Proof: TODO $\square$

Permutations of a Multiset

Theorem 6: Let S be a finite multiset with k different types and repetition numbers $r_1, \ldots, r_k$. Let the size of S be $n = r_1 + \ldots + r_k$. Then the number of n-permutations of S equals

$$\binom{n}{r_1, \ldots, r_k}$$

Proof: In an n-permutation there are n positions that need to be assigned a type.

First, choose the r_1 positions for the first type, there are $\binom{n}{r_1}$ ways to do so. Then, assign r_2 positions for the second type, out of the $(n - r_1)$ positions that are still available, there are $\binom{n-r_1}{r_2}$ ways to do so. Continue for all k types. The total number of choices will be:

$$\binom{n}{r_1} \cdot \binom{n - r_1}{r_2} \cdot \ldots \cdot \binom{n - r_1 - r_2 - \ldots - r_{k-1}}{r_k} = \binom{n}{r_1, \ldots, r_k}$$

$\square$

k-Combinations of an *Infinite* Multiset

Example: Suppose you have a *sufficiently large* amount of each type of fruit (🍌, 🍎, 🍐) in the supermarket, and you want to buy *two* fruits. How many choices do you have?

There are exactly *six* combinations: {🍌, 🍌}, {🍌, 🍎}, {🍌, 🍐}, {🍎, 🍎}, {🍎, 🍐}, {🍐, 🍐}.

Note that your selection is *not ordered*, so {🍐, 🍎} and {🍎, 🍐} are considered the *same* choice.

k-Combinations of an *Infinite* Multiset [2]

Theorem 7: Let $k, s \in \mathbb{N}$ and let S be a multiset with s types and large repetition numbers (each $r_1, \ldots, r_s$ is *at least k*), then the number of k-combinations of S equals

$$\binom{k+s-1}{k} = \binom{k+s-1}{s-1}$$

Proof: Let $S = \{\infty\text{🍌}, \infty\text{🍎}, \infty\text{🍐}\}$, so $s = 3$.

- Suppose $k = 5$.
- Consider a 5-combination of S: {🍌, 🍎, 🍌, 🍐, 🍐}.
- Reorder and group: {🍌 🍌 | 🍎 | 🍐 🍐}.
- Convert to *dots* and *bars*: $\bullet\bullet \mid \bullet \mid \bullet\bullet$
- Represent as a multiset: $M = \{k \cdot \bullet, (s-1) \cdot \mid\}$
- Observe: each *permutation* of k dots and $(s-1)$ bars corresponds to a *k-combination* of S.
- Permute the 2-type multiset: $\binom{k+s-1}{k,s-1}$ ways, by <u>Theorem 5</u>.

□

§5 Compositions

Weak Compositions

Definition 16: A *weak composition* of a non-negative integer $k \geq 0$ into s parts is a *solution* to the equation $b_1 + \ldots + b_s = k$, where each $b_i \geq 0$.

Example: Let $k = 5$, $s = 3$. Possible non-negative integer solutions for $b_1 + b_2 + b_3 = 5$ are:

- $(b_1, b_2, b_3) = (1, 1, 3)$
- $(b_1, b_2, b_3) = (1, 3, 1)$
- $(b_1, b_2, b_3) = (2, 0, 3)$
- $(b_1, b_2, b_3) = (0, 5, 0)$
- ... (total 21 solutions)

Note: If M is a multiset over groundset $\{1, \ldots, s\}$ with all multiplicities infinite ($r_i = \infty$), then for $k \geq 0$, the number of sub-multisets of M of size k is exactly the number of weak compositions of k into s parts.

Counting Weak Compositions

Theorem 8: There are $\binom{k+s-1}{k,s-1}$ *weak compositions* of $k > 0$ into s parts.

Proof: Observe that $k = \overbrace{\underbrace{1+1}_{b_1} + \underbrace{\ldots}_{b_i} + \underbrace{1+1}_{b_s}}^{k \text{ ones}}$.

Use the *stars-and-bars* method to count the number of s groups composed of k "ones". $\square$

Example: Let $k = 3$. There are $\binom{3+3-1}{3,3-1} = \binom{5}{3} = \binom{5}{2} = 10$ ways to decompose $k = 3$ into $s = 3$ parts:

$$\begin{aligned} k = 3 &= \\ &= 0+1+2 = 0+2+1 \\ &= 1+0+2 = 1+2+0 = 1+1+1 \\ &= 2+0+1 = 2+1+0 \\ &= 3+0+0 = 0+3+0 = 0+0+3 \end{aligned}$$

Compositions

Definition 17: A *composition* of a positive integer $k \geq 1$ into s *positive* parts is a *solution* to the equation $b_1 + \ldots + b_s = k$, where each $b_i > 0$.

Theorem 9: There are $\binom{k-1}{s-1}$ *compositions* of $k > 0$ into s positive parts.

Theorem 10: The total number of compositions of $k > 0$ into *some* number of positive parts is

$$\sum_{s=1}^{k} \binom{k-1}{s-1} = 2^{k-1}$$

Parallel Summation Identity

Q: How many integer solutions are there to the *inequality* $b_1 + \ldots + b_s \leq k$, where each $b_i \geq 0$?

Theorem 11: $\displaystyle\sum_{m=0}^{k} \binom{m+s-1}{m} = \binom{k+s}{k}$

Proof *(hint)*: Introduce a "dummy" variable b_{s+1} to take up the *slack* between $b_1 + \ldots + b_s$ and k. Construct a bijection with the solutions to $b_1 + \ldots + b_s + b_{s+1} = k$, where $b_i \geq 0$. $\square$

§6 Set Partitions

Set Partitions

Definition 18: A *partition* of a set X is a set of non-empty subsets of X such that every element of X belongs to exactly one of these subsets.

Equivalently, a family of sets P is a partition of X iff:

1. The family P does not contain the empty set: $\emptyset \notin P$.
2. The union of P is X, that is, $\bigcup_{A \in P} A = X$. The sets in P are said to *cover* X.
3. The intersection of any two distinct sets in P is empty: $\forall A, B \in P. (A \neq B) \to (A \cap B = \emptyset)$.
   The sets in P are said to be *pairwise disjoint* or *mutually exclusive*.

The sets in P are called *blocks*, *parts*, or *cells*, of the partition.

The block in P containing an element $x \in X$ is denoted by $[x]$.

Examples of Set Partitions

Example: The empty set $X = \emptyset$ has exactly one partition, $P = \emptyset$.

Example: Any singleton set $X = \{x\}$ has exactly one partition, $P = \{\{x\}\}$.

Example: For any non-empty proper subset $A \subset U$, the set A and its complement form a partition of U, namely $P = \{A, U - A\}$.

Example: The set $X = \{1, 2, 3\}$ has five partitions:

1. $\{\{1\}, \{2\}, \{3\}\}$ or $1 \mid 2 \mid 3$
2. $\{\{1\}, \{2, 3\}\}$ or $1 \mid 2\ 3$
3. $\{\{1, 2\}, \{3\}\}$ or $1\ 2 \mid 3$
4. $\{\{1, 3\}, \{2\}\}$ or $1\ 3 \mid 2$
5. $\{\{1, 2, 3\}\}$ or $1\ 2\ 3$

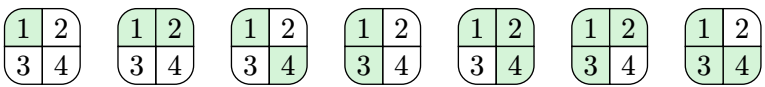
Example: The following are *not* partitions of $\{1, 2, 3\}$:

- $\{\{\}, \{1, 3\}, \{2\}\}$, because it contains the empty set.
- $\{\{1, 2\}, \{2, 3\}\}$, because the element 2 is contained in more than one block.
- $\{\{1\}, \{3\}\}$, because no block contains the element 3.

Counting Set Partitions

Definition 19: The number of partitions of a set X (of size $n = |X|$) into k non-empty blocks ("unlabeled subsets") is called a *Stirling number of the second kind* and denoted $S(n, k)$ or $\left\{ {n \atop k} \right\}$.

Example: Let $X = \{1, 2, 3, 4\}$, $k = 2$. There are 7 possible partitions:



Theorem 12: Let $\left\{ {n \atop 0} \right\} = 0$ for $n \geq 1$, $\left\{ {0 \atop k} \right\} = 0$ for $k \geq 1$, and $\left\{ {0 \atop 0} \right\} = 1$. For $n, k \geq 1$, we have:

$$\left\{ {n \atop k} \right\} = \left\{ {n-1 \atop k-1} \right\} + k \cdot \left\{ {n-1 \atop k} \right\}$$

Proof *(informal)*: TODO $\square$

Bell Numbers

Definition 20: The total number of partitions of a set X of size $n = |X|$ (into an arbitrary number of non-empty blocks) is called a *Bell number* and denoted B_n.

$$B_n = \sum_{k=0}^{n} \left\{ {n \atop k} \right\}$$

Note: Consider the special case of $n = 0$. There is exactly *one* partition of $\emptyset$ into non-empty parts: $\emptyset = \bigcup_{A \in \emptyset} A \in \emptyset$. Every $A \in \emptyset$ is non-empty, since no such A exists. Thus, we have $B_0 = S(0, 0) = 1$.

Bell Numbers [2]

Theorem 13: For $n \geq 1$, we have the recursive identity for Bell numbers:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

Proof: Every partition of $[n]$ has one part that contains the number n. In addition to n, this part also contains k other numbers (for some $0 \leq k \leq n-1$). The remaining $n-1-k$ elements are partitioned arbitrarily. From this correspondence, we obtain the desired identity:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{n-1-k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

$\square$

§7 Integer Partitions

Integer Partitions

Definition 21: An *integer partition* of a positive integer $n \geq 1$ into k *positive* parts is a *solution* to the equation $n = a_1 + \ldots + a_k$, where $a_1 \geq a_2 \geq \ldots \geq a_k \geq 1$.

- The number of integer partitions of n into k positive non-decreasing parts is denoted $p_k(n)$ and defined recursively:

$$p_k(n) = \begin{cases} 0 & \text{if } k > n \\ 0 & \text{if } n \geq 1 \text{ and } k = 0 \\ 1 & \text{if } n = k = 0 \\ p_k(n-k) + p_{k-1}(n-1) & \text{if } 1 \leq k \leq n \end{cases}$$

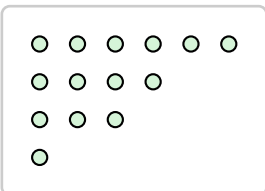
- The number of partitions of n (into an arbitrary number of parts) is the *partition function* $p(n)$:

$$p(n) = \sum_{k=0}^{n} p_k(n)$$

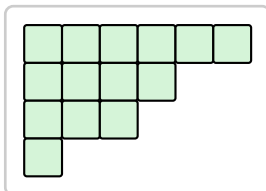
Ferrer Diagrams and Yound Tableaux

Example: Consider an integer partition: $14 = 6 + 4 + 3 + 1$.

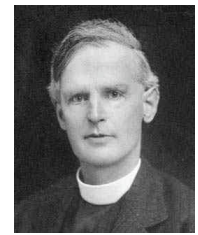
Ferrer Diagram



Young Tableaux



Norman Ferrer



Alfred Young

§8 Inclusion–Exclusion

The Inclusion–Exclusion Principle

TODO: small example of PIE with 2 or 3 sets

Principle of Inclusion–Exclusion (PIE)

Theorem 14: Let X be a finite set and $P_1, \ldots, P_m$ *properties*.

- Define $X_i = \{x \in X \mid x \text{ has } P_i\}$, i.e. the set of all elements from X having a property P_i.
- Define for $S \subseteq [m]$ the set $N(S) = \{x \in X \mid \forall i \in S : x \text{ has } P_i\}$. Observe: $N(S) = \bigcap_{i \in S} X_i$.

The number of elements of X that satisfy *none* of the properties $P_1, \ldots, P_m$ is given by

$$|X \setminus (X_1 \cup \ldots \cup X_m)| = \sum_{S \subseteq [m]} (-1)^{|S|} |N(S)| \tag{1}$$

Proof: Consider any $x \in X$. If $x \in X$ has none of the properties, then $x \in N(\emptyset)$ and $x \notin N(S)$ for any other $S \neq \emptyset$. Hence *x contributes 1* to the sum (1).

If $x \in X$ has exactly $k \geq 1$ of the properties, call this set $T \in \binom{[m]}{k}$. Then $x \in N(S)$ iff $S \subseteq T$.
The *contribution of x* to the sum (1) is $\sum_{S \subseteq T} (-1)^{|S|} = \sum_{i=0}^{k} \binom{k}{i} (-1)^i = 0$, i.e. *zero*. $\square$

Note: In the last step, we used that for any $y \in \mathbb{R}$ we have $(1-y)^k = \sum_{i=0}^{k} \binom{k}{i} (-y)^i$ which implies (for $y = 1$) that $0 = \sum_{i=0}^{k} \binom{k}{i} (-1)^i$.

Very Useful Corollary of PIE

Corollary 14.1: 🙀

$$\left|\bigcup_{i\in[m]} X_i\right| = |X| - \sum_{S\subseteq[m]} (-1)^{|S|}|N(S)| = \sum_{\emptyset \neq S\subseteq[m]} (-1)^{|S|-1}|N(S)|$$

Applications of PIE

Let's state the principle of inclusion-exclusion using a rigid pattern:

1. *Define "bad" properties.*

   Identify the things to count as the elements of some universe X except for the whose having *at least one* of the "bad" properties $P_1, \ldots, P_m$. In other words, we want to count $X \setminus (X_1 \cup \ldots \cup X_m)$.

2. *Count $N(S)$.*

   For each $S \subseteq [m]$, determine $N(S)$, the number of elements of X having *all* bad properties P_i for $i \in S$.

3. *Apply PIE.*

   Use <u>Theorem 14</u> to obtain a closed formula for $|X \setminus (X_1 \cup \ldots \cup X_m)|$.

Counting Surjections via PIE

Theorem 15: The number of surjections from $[k]$ to $[n]$ is given by

$$\left|\left\{f : [k] \underset{\text{surj.}}{\longrightarrow} [n]\right\}\right| = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k$$

Proof: Let X be the set of all maps from $[k]$ to $[n]$.

1. *Define bad properties:* Define the "bad" property P_i for $i \in [n]$ as "i is not in the image of f", i.e.

$$f : [k] \longrightarrow [n] \text{ has property } P_i \quad \leftrightarrow \quad \forall j \in [k] : f(j) \neq i$$

   The *surjective* functions are exactly those functions that *do not* have any of the "bad" properties.

2. *Count $N(S)$:* We claim $N(S) = (n - |S|)^k$ for any $S \subseteq [n]$. To see this, observe that f has all properties with indices from S if and only if $f(i) \notin S$ for all $i \in [k]$. In other words, f must be a function from $[k]$ to $[n] \setminus S$, and there are $(n - |S|)^k$ of those.

Counting Surjections via PIE [2]

3. *Apply PIE:* Using Theorem 14, the number of surjections from $[k]$ to $[n]$ is

$$\begin{aligned}|X \setminus (X_1 \cup \ldots \cup X_n)| &\overset{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\ &= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)^k \\ &= \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k\end{aligned}$$

In the last step, we used that $(-1)^{|S|}(n - |S|)^k$ only depends on the size of S, and there are $\binom{n}{i}$ sets $S \subseteq [n]$ of size i.

$\square$

More Useful Corollaries

Corollary 15.1: Consider the case $n = k$. A function from $[n]$ to $[n]$ is a *surjection* iff it is a *bijection*. Since there are $n!$ bijections on $[n]$ (namely, all permutations), we have the following identity:

$$n! = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^i$$

Corollary 15.2: A surjection from $[k]$ to $[n]$ can be seen as a partition of $[k]$ into n non-empty distinguishable (labeled) parts (the map assigns a part to each $i \in [k]$).

Since the partition of $[k]$ into n non-empty indistinguishable parts is denoted $s_n^{\Pi}(k)$, and there are $n!$ ways to assign labels to n parts, we obtain that the number of surjections is equal to $n! s_n^{\Pi}(k)$:

$$n! s_n^{\Pi}(k) = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)^k$$

Derangements

Theorem 16: The *derangements* D_n on n elements are permutations of $[n]$ without fixed points.

The number of derangements is given by

$$|D_n| = \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)!$$

Proof: Let X be the set of all permutations of $[n]$.

1. Define the "bad" property P_i to mean "π has a fixpoint i" ($i \in [n]$):

$$\pi \in X \text{ has property } P_i \quad \leftrightarrow \quad \pi(i) = i$$

2. We claim $N(S) = (n - |S|)!$ for any $S \subseteq [n]$.

   Indeed, $\pi \in X$ has all properties with indices from S if and only if all $i \in S$ are fixed points of π. On the other elements, i.e. on $[n] \setminus S$, π may be an arbitrary bijection, so there are $(n - |S|)!$ choices for π.

Derangements [2]

3. Using Theorem 14, the number of derangements is given by

$$
\begin{aligned}
|X \setminus (X_1 \cup \ldots \cup X_n)| &\overset{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\
&= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)! \\
&= \sum_{i=0}^{n} (-1)^i \binom{n}{i} (n-i)!
\end{aligned}
$$

In the last step, we used that $(-1)^{|S|}(n - |S|)!$ only depends on the size of S, and there are $\binom{n}{i}$ sets $S \subseteq [n]$ of size i.

$\square$

§9 Generating Functions

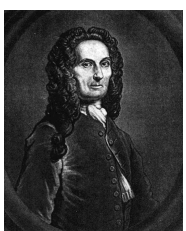
Generating Functions

A generating function is a device somewhat similar to a bag. Instead of carrying many little objects detachedly, which could be embarrassing, we put them all in a bag, and then we have only one object to carry, the bag.

— George Pólya, Mathematics and Plausible Reasoning [1]

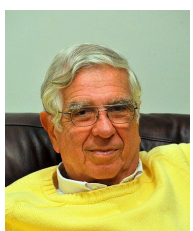
A generating function is a clothesline on which we hang up a sequence of numbers for display.

— Herbert Wilf, generatingfunctionology [2]



Abraham de Moivre

George Pólya



Herbert Wilf

Ordinary Generating Functions

Definition 22: An *ordinary generating function* (OGF) of a sequence a_n is a *power series*

$$G(a_n; x) = \sum_{n=0}^{\infty} a_n x^n$$

Example: The *sequence* $a_n = (a_0, a_1, a_2, \ldots)$ is *generated* by the OGF $G(x) = a_0 + a_1 x + a_2 x^2 + \ldots$

Example: $G(x) = 3 + 8x^2 + x^3 + \frac{1}{7}x^5 + 100x^6 + \ldots$ *generates* the sequence $\left(3, 8, 1, \frac{1}{7}, 100, \ldots\right)$

Example: Consider a long division of 1 by $(1 - x)$, the result is an infinite power series

$$\frac{1}{1-x} = 1 + x^1 + x^2 + x^3 + \ldots = \sum_{n=0}^{\infty} x^n$$

Note that all coefficients are 1. Thus, the generating function of $(1, 1, 1, \ldots)$ is $G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$.

More Examples of Generating Functions

Another proof that $(1, 1, 1, \ldots)$ is generated by $G(x) = 1 + x + x^2 + x^3 + \ldots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = S$:

$$\begin{array}{rcl} S & = & 1 + x + x^2 + x^3 + \ldots \\ x \cdot S & = & \phantom{1 + {}} x + x^2 + x^3 + \ldots \\ \hline S - x \cdot S & = & 1 \end{array}$$

Thus, $S = \dfrac{1}{1-x}$

More Examples of Generating Functions [2]

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \ldots \quad \text{generates } (1, 1, 1, \ldots) \quad \text{(constant 1)}$$

$$\frac{2}{1-x} = \sum_{n=0}^{\infty} 2x^n = 2 + 2x + 2x^2 + 2x^3 + \ldots \quad \text{generates } (2, 2, 2, \ldots) \quad \text{(constant 2)}$$

$$\frac{x}{1-x} = \sum_{n=1}^{\infty} x^n = x + x^2 + x^3 + \ldots \quad \text{generates } (0, 1, 1, 1, \ldots) \quad \text{(right shift)}$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n = 0 + 1 - x + x^2 - x^3 + \ldots \quad \text{generates } (1, -1, 1, \ldots) \quad \text{(sign-alternating 1's)}$$

$$\frac{1}{1-3x} = \sum_{n=0}^{\infty} 3^n x^n = 1 + 3x + 9x^2 + 27x^3 + \ldots \quad \text{generates } (1, 3, 9, \ldots) \quad \text{(powers of 3)}$$

$$\frac{1}{1-x^2} = \sum_{n=0}^{\infty} x^n = 1 + x^2 + x^4 + x^6 + \ldots \quad \text{generates } (1, 0, 1, 0, \ldots) \quad \text{(regular gaps)}$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty} (n+1)x^n = 0 + x + 2x^2 + 3x^3 + \ldots \quad \text{generates } (1, 0, 1, 0, \ldots) \quad \text{(natural numbers)}$$

More Examples of Generating Functions [3]

$$\frac{1-x^{n+1}}{1-x} = \frac{1}{1-x} - \frac{x^{n+1}}{1-x} =$$

$$\triangleq (1,1,1,...) - (\underbrace{0,0,...,0}_{n+1 \text{ zeros}},1,1,...) =$$

$$= (\underbrace{1,1,...,1}_{n+1 \text{ ones}},0,0,...) =$$

$$\triangleq 1 + x + x^2 + ... + x^n$$

Exercises

Example: Find GF for odd numbers: $(1, 3, 5, \ldots)$.

Example: Find GF for $(1, 3, 7, 15, 31, 63)$, which satisfies $a_n = 3a_{n-1} - 2a_{n-2}$ with $a_0 = 1$, $a_1 = 3$.

Solving Combinatorial Problems via Generating Functions

Example: Find the number of integer solutions to $x_1 + x_2 + x_3 = 12$ with $0 \leq x_i \leq 6$.

- Possible values for x_1: $0, 1, \ldots, 6$.
- Write this as a polynomial $x^0 + x^1 + \ldots + x^6$.
- Do the same for x_2 and x_3.
- Since all combinations of x_1, x_2 and x_3 are suitable non-conflicting solutions, we can multiply those polynomials and obtain the *generating function* $G(x) = \left(1 + x + x^2 + \ldots + x^6\right)^3$.
- For each n, the coefficient of x^n in $G(x)$ is the number of integer solutions to $x_1 + x_2 + x_3 = n$.
- Specifically, we are interested in the coefficient of x^{12} in $G(x)$, denoted $[x^{12}]G(x)$.
- Use ~~pen and paper~~ Wolfram Alpha to expand $G(x)$:

$$G(x) = x^{18} + 3x^{17} + 6x^{16} + \ldots + \underbrace{28x^{12}} + \ldots + 6x^2 + 3x + 1$$

- The answer is $[x^{12}]G(x) = 28$ solutions.

Slightly More Complex Combinatorial Problem

Example: Suppose we have marbles of three different colors (🔴, 🟢, 🔵), and we want to *count* the number of ways to select 20 marbles, such that:

- There are an even number of 🔴 : $1 + x^2 + x^4 + \ldots + x^{20}$.
- There are at least 12 🟢 : $x^{12} + x^{13} + \ldots + x^{20}$.
- There are at most 5 🔵 : $1 + x + x^2 + x^3 + x^4 + x^5$.

Multiply polynomials and find $[x^{20}]G(x)$:

$$
\begin{aligned}
&[x^{20}](1 + x^2 + x^4 + \ldots + x^{20})(x^{12} + x^{13} + \ldots + x^{20})(1 + x + x^2 + x^3 + x^4 + x^5) = \\
&= [x^{20}](x^{45} + 2x^{44} + \ldots + 21x^{20} + \ldots + 2x^{13} + x^{12}) \\
&= 21
\end{aligned}
$$

Using Power Series in Combinatorial Problems

Example: Find the number of integer solutions to $x_1 + x_2 + x_3 = 12$ with $x_1 \geq 2$, $3 \leq x_2 \leq 6$, $x_3 \leq 9$.

- Compose the generating function:

$$G(x) = (x^2 + x^3 + \ldots) \cdot (x^3 + x^4 + x^5 + x^6) \cdot (1 + x + x^2 + \ldots + x^9)$$

- Substitute the power series with the corresponding simple forms:

$$G(x) = \left(x^2 \cdot \frac{1}{1-x}\right) \cdot \left(x^3 \cdot \frac{1-x^4}{1-x}\right) \cdot \left(\frac{1-x^{10}}{1-x}\right)$$

Bibliography

[1] G. Pólya, *Mathematics and Plausible Reasoning, Volume 1: Induction and Analogy in Mathematics*. 1954.

[2] H. S. Wilf, *generatingfunctionology*, 3rd ed. 2006.